



Vijayaragavan S 251801381-2025-AIDS ▾

V2

Started on	Thursday, 17 September 2026, 8:14 AM
State	Finished
Completed on	Thursday, 17 September 2026, 8:20 AM
Time taken	6 mins 5 secs
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 10.00 out of 10.00**Playing with Numbers:**

Ram and Sita are playing with numbers by giving puzzles to each other. Now it was Ram term, so he gave Sita a positive integer 'n' and two numbers 1 and 3. He asked her to find the possible ways by which the number n can be represented using 1 and 3. Write any efficient algorithm to find the possible ways.

Example 1:

Input: 6

Output: 6

Explanation: There are 6 ways to 6 represent number with 1 and 3

1+1+1+1+1+1

3+3

1+1+1+3

1+1+3+1

1+3+1+1

3+1+1+1

Input Format

First Line contains the number n

Output Format

Print: The number of possible ways 'n' can be represented using 1 and 3

Sample Input

6

Sample Output

6

Answer: (penalty regime: 0 %)

```
1 #include <stdio.h>
2 long long countWays(int n) {
3     if (n == 0 || n == 1 || n == 2) {
4         return 1;
5     }
6     long long prev3 = 1;
7     long long prev2 = 1;
8     long long prev1 = 1;
9     long long current = 0;
10    for (int i = 3; i <= n; i++) {
11        current = prev1 + prev3;
12        prev3 = prev2;
13        prev2 = prev1;
14        prev1 = current;
15    }
16    return current;
17 }
18 int main() {
19     int n;
20     if (scanf("%d", &n) != 1) {
21         return 1;
22     }
23     printf("%lld\n", countWays(n));
24     return 0;
25 }
```

	Input	Expected	Got	
✓	6	6	6	✓
✓	25	8641	8641	✓
✓	100	24382819596721629	24382819596721629	✓

Passed all tests! ✓

Correct

Marks for this submission: 10.00/10.00.

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